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BSIT 2H-G1

JAVA LINKED LIST SAMPLE PROBLEMS

General Learning Objectives

After completing these activities, students should be able to:

1. Create a LinkedList using java.util.LinkedList.
2. Add elements to a LinkedList.
3. Insert elements at a specific position.
4. Access elements using get().
5. Update elements using set().
6. Search using contains().
7. Remove elements.
8. Display the contents of a LinkedList.
9. Explain what happened to the list after each operation.

GENERAL INSTRUCTIONS

For **each problem**, follow these rules:

1. Import LinkedList

Your program must use: import java.util.LinkedList;

2. Use your name in the class name

The class name must follow this format: LinkedListProb_FirstnameLastname

For example, if the student's name is **Juan Dela Cruz**: LinkedListProb_JuanDelaCruz.java **3. File name**

The Java file must have the same name as the class.

4. Use comments

Students should add comments to identify important operations.

5. Submit

For each problem, students should submit:

- .java source code
- Screenshot of the program
- Screenshot of the output
- Answers to the reflective questions

SAMPLE PROBLEM 1

Create a Java program that uses a LinkedList<String> to store the names of five students.

The program should:

1. Add five student names.
2. Display the original list.
3. Add another student's name at the beginning.
4. Add another student's name at the end.
5. Display the updated list.
6. Search for a particular student's name.
7. Display the number of students in the list.

Expected Output:

```
Original Student List:
[Ana, Ben, Carlo, Diana, Edward]

Updated Student List:
[Maria, Ana, Ben, Carlo, Diana, Edward, Felix]

Carlo is found in the list.
Number of students: 7
```

SAMPLE PROBLEM 2

Product Inventory Problem

Create a Java program that uses a LinkedList<String> to manage products in a small computer store.

The program should:

1. Add five products.
2. Display the original product list.
3. Insert a new product at index 2.
4. Update the product at index 3.
5. Search for a product.
6. Remove one product.
7. Display the final product list

Starting Products

Use: Keyboard, Mouse, Monitor, Printer, Webcam

Perform the following operations:

Operation 1 Insert:

USB Hub

at index 2.

Operation 2

Change the item at index 3 to:

Mechanical Keyboard

Operation 3 Search

for:

Printer

Operation 4 Remove:

Mouse

Expected Output:

```
Original Product List:
[Keyboard, Mouse, Monitor, Printer, Webcam]

After inserting USB Hub:
[Keyboard, Mouse, USB Hub, Monitor, Printer, Webcam]

After updating index 3:
[Keyboard, Mouse, USB Hub, Mechanical Keyboard, Printer, Webcam]

Printer is available.

Final Product List:
[Keyboard, USB Hub, Mechanical Keyboard, Printer, Webcam]
```

SAMPLE PROBLEM 3

Simple Course Enrollment System

Difficulty: Intermediate Beginner

This problem combines more LinkedList methods.

Problem

Create a Java program that manages students enrolled in a course.

The program should:

1. Create a LinkedList of students.
2. Add five students.
3. Display the students.
4. Add a new student at the beginning.
5. Add another student at the end.
6. Insert another student at index 3.
7. Display the student at index 2.
8. Search for a particular student.
9. Remove one student.
10. Display the final list.
11. Display the total number of students.
12. Check if the list is empty.

Students

Start with:

Juan
Maria
Pedro
Anna
Luis

Then perform:

Add beginning:

Carlo

Add end:

Sofia

Insert at index 3:

Daniel

Get index 2

Search for

Maria

Remove

Pedro

Expected Output:

```
Original Student List:
[Juan, Maria, Pedro, Anna, Luis]

After adding Carlo and Sofia:
[Carlo, Juan, Maria, Pedro, Anna, Luis, Sofia]

After inserting Daniel at index 3:
[Carlo, Juan, Maria, Daniel, Pedro, Anna, Luis, Sofia]

Student at index 2: Maria
Maria is enrolled.

After removing Pedro:
[Carlo, Juan, Maria, Daniel, Anna, Luis, Sofia]

Total students: 7
The student list is not empty.
```

Student Reflection

After completing the three problems, students should answer the following questions in their own words.

Reflective Questions

1. What is the purpose of: import java.util.LinkedList; ?
   We import java.util.LinkedList to allow us to use special methods essential for creating and managing a LinkedList.

2. What is the difference between: add() and: addFirst() ?
   add() adds a new element following the last existing sequence. On the other hand, **addFirst()** let us add an element directly at the beginning of the LinkedList

3. What happens when you use: addLast() ?
   If we use **addLast(),** we can add an element directly at the end of the LinkedList

4. What is the purpose of: get() ? Give one example.
   Using **get()** allows us to access a specific value of an element based on a specific index. For instance, in a LinkedList of *studentNames,* we can get the name of the student on a specific index. students.get(2) would give us the student name stored on index number 2.

5. What is the difference between: set() and: add() ?
   set() allows us to update or change a specific element, while **add** allows us to add another element in the list

6. What does: contains() do?
   contains() allows us to tell if a specific value is stored or is existing within the list

7. What does: size() tell us about a LinkedList?
   size() tells us the numbers of elements stored in the list

8. Why is it important to remember that Java uses **zero-based indexing**?
   It is important to remember that Java index always starts with 0

9. Which LinkedList method did you find easiest to understand? Explain why.
   I find **add()** easiest to understand as it is similar to ArrayList which is somewhat familiar with us

10. Which LinkedList operation was the most confusing for you? Explain what made it difficult.
    I find each of the operations I used equally difficult but easy to understand and implement within the project.

11. If the list contains: [10, 20, 30, 40] what will happen after: numbers.add(2, 25); Explain your answer.

    The value 25 would be added or inserted to index number 2. Making the list {10, 20, 25, 30, 40}

12. If you have: [10, 20, 30, 40] what will happen after: numbers.remove(Integer.valueOf(30)); Explain your answer.
    The value of 30 would be removed from the list. Making the LinkedList {10, 20, 40}

Java LinkedList Activity Rubric — 100 Points

Criteria	Excellent	Good	Developing	Beginning	Points
Program Requirements	All required operations are implemented correctly	Most requirements are implemented	Several requirements are missing	Major requirements are missing	25
Correct Use of LinkedList Methods	Correctly uses methods such as add(), addFirst(), addLast(), get(), set(), contains(), remove(), size() and isEmpty()	Minor errors in method usage	Several method errors	Methods are largely incorrect or missing	25

Program Output	Output is complete and matches the required operations	Minor output issues	Some output is missing or incorrect	Output is largely incorrect	20
Code Organization	Code is properly structured, readable, and includes useful comments	Generally organized with minor issues	Some organization problems	Difficult to follow	10
Naming Convention	Correct class and file naming convention using student's name	Minor naming error	Several naming issues	Does not follow required naming convention	5
Reflection	Answers are clear, complete, and demonstrate understanding	Answers show adequate understanding	Answers are incomplete	Answers show little understanding	10
Program Demonstration / Screenshot	Complete screenshots of code and output are provided	One minor missing item	Several screenshots are missing	No proper evidence provided	5
TOTAL					100